

Chapter 64. Inhaled Antibiotic Therapy

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Inhaled Antibiotic Therapy: Introduction

The incidence of ventilator-associated pneumonia ranges between 8% and 28% in patients receiving mechanical ventilation for more than 48 hours, and between 34% and 70% in patients with acute lung injury or acute respiratory distress syndrome.¹ It prolongs the duration of stay in the intensive care unit and hospital, and increases costs.² Associated mortality ranges from 24% to 76%, and appears far greater than the mortality resulting from other nosocomial infections. It may even exceed 85% when high-risk gram-negative bacteria, such as *Pseudomonas aeruginosa* or *Acinetobacter baumannii*, are the causative pathogens.³ Many studies demonstrate that early intravenous administration of appropriate antibiotics improves the prognosis. Lung deposition of antibiotics, however, administered by the intravenous route is either limited or poorly documented and treatment failure is common, leading to increased dosage and risk of systemic toxicity. Despite antimicrobial therapy and adequate supporting treatment, the mortality rate from ventilator-associated pneumonia remains high, indicating a need for a more effective route of administration. Inhaled antibiotic therapy may represent such an alternative.

Rationale

At least three major theoretical arguments support the administration of inhaled antibiotics in critically ill patients with ventilator-associated pneumonia: the very pathogenesis of lung infection that originates in the tracheobronchial tree, the possibility of obtaining high lung-tissue concentrations by bypassing the alveolar-capillary barrier, and the potential for decreasing systemic toxicity.

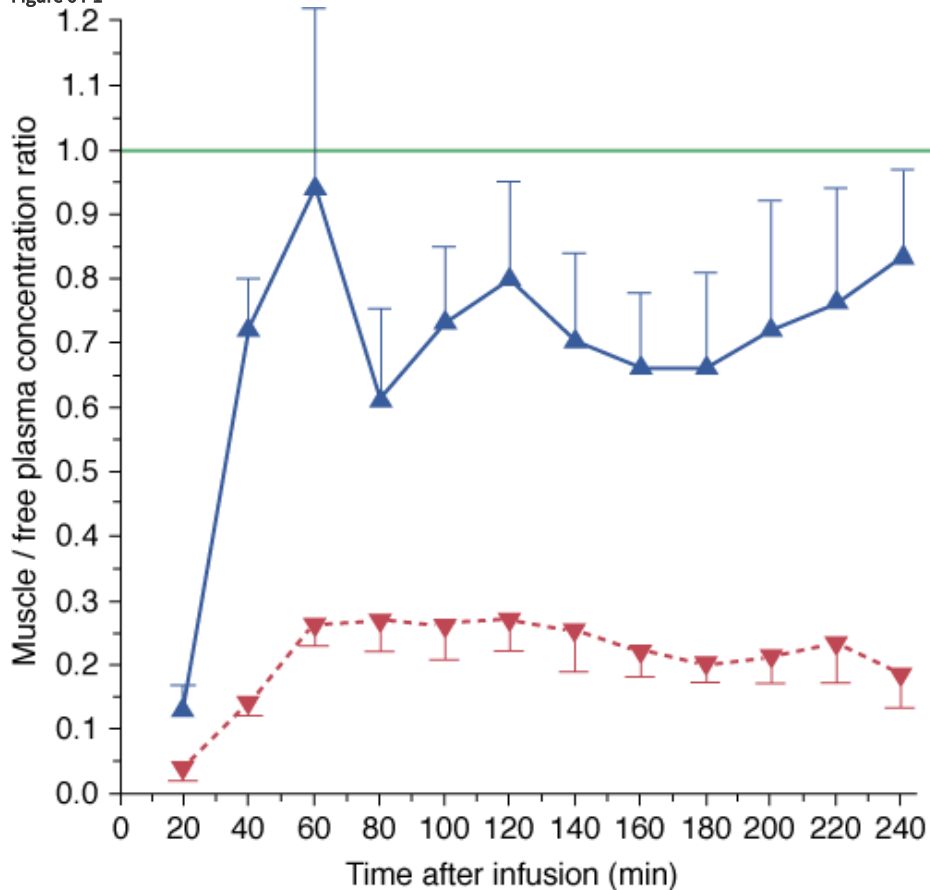
Deep Lung Infection Originates in the Upper Airways

The normal human respiratory tract possesses efficient defenses against bacteria colonizing the pharynx. The glottis and larynx serve as natural anatomic barriers. The cough reflex, mucociliary clearance, and regional immunity contribute to elimination of invading pathogens and prevention of infection deep in the lungs. Endotracheal and tracheostomy tubes bypass the natural barrier between the oropharynx and tracheobronchial tree. Bacteria penetrate into the trachea by leakage of infected secretions and/or contaminated gastric contents around the low-pressure cuff of an endotracheal tube.⁴ In ventilated patients, deep sedation depresses the cough reflex whereas endotracheal intubation inhibits the ciliary escalator.^{5,6} In addition, the internal wall of the endotracheal tube rapidly becomes coated with an antibiotic-resistant bacterial biofilm, which then can become fragmented and disseminated into the deep lung during tracheal suctioning or fiber-optic procedures.^{7,8} Consequently, it appears reasonable to hypothesize that antibiotics administered by the inhalational route may reduce bacterial inoculum by stopping the continuous bacterial seeding from the upper airways.

Bypassing the Alveolar–Capillary Barrier May Provide High Antibiotic Tissue Concentrations

Killing bacteria infecting the lung parenchyma requires that an antibiotic achieve a pulmonary concentration at least five times greater than minimal inhibitory concentrations for the infecting pathogen. Reaching concentrations lower than mutant prevention concentrations in the infected parenchyma may trigger the emergence of resistant bacterial strains.⁹ When antibiotics are intravenously administered, the alveolar–capillary barrier imposes a difficult-to-cross obstacle, which impairs lung deposition even if lung inflammation increases capillary permeability. Pulmonary vasoconstriction and regional thrombosis, two pathophysiologic abnormalities characterizing severe bronchopneumonia, reduce lung perfusion and tend to impair pulmonary penetration of circulating antibiotics. Major surgery and septic shock are associated with a significant decrease in the ratio between systemic and interstitial antibiotic concentrations (Fig. 64-1). All these factors contribute to markedly reduce the antibiotic concentrations at the site of infection, and may explain failure of intravenous antibacterial treatment.^{10,11} During the initial phase of pulmonary infection, some degree of aeration persists within the bronchiolar lumen and alveolar space. Therefore, the early inhalation of antibiotics may represent a unique opportunity to reach bactericidal concentrations at the site of infection. In the presence of severe bronchopneumonia, the existence of the alveolar–capillary barrier and reduced lung perfusion are far from deleterious; instead, they may be beneficial by limiting systemic absorption and reducing systemic toxicity of nebulized antibiotics.

Figure 64-1



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Time course of the ratio between unbound piperacillin concentrations measured in the interstitial space fluid of skeletal muscle and the concentrations measured in the plasma (mean $\pm$ standard error of mean) after administration of a single intravenous dose of 4 g to six septic patients (*inverted triangle dotted line*) and six control healthy

volunteers (*triangle solid line*). Piperacillin levels were measured using high-pressure liquid chromatography via microdialysis probes inserted into skeletal muscles. (Used, with permission, from Joukhadar et al.¹¹)

Factors Influencing Lung Deposition

A number of factors influence lung deposition of nebulized antibiotics during mechanical ventilation: aerosol particle size, type of nebulizer, ventilator settings, grade of bronchopneumonia, and lung aeration. A better understanding of the conditions regulating lung deposition and the tremendous growth of technological innovations¹² potentially make the nebulization of antibiotics an attractive alternative to the classic intravenous administration.

Aerosol Particle Size

An essential condition for an aerosol particle to reach the deep lung is to ensure that the mass median aerodynamic diameter ranges between 1 and 5 μm .¹³ Larger particles tend to impact and attach to ventilator circuits and endotracheal tubes, thereby limiting lung deposition.^{14,15} Aerosol-generating devices, such as dry powder inhalers, metered-dose inhalers, jet nebulizers, ultrasonic nebulizers, or vibrating plate nebulizers, all produce particles whose mass median aerodynamic diameter is less than 5 μm . It has to be emphasized that nebulizers that produce the smallest particles require considerably more time to deliver a standard antibiotic dose.¹⁶

Type and Position of Nebulizers

A nebulizer equipped with a large reservoir is needed for delivering sufficient amounts of antibiotics to the respiratory system. As a consequence, metered-dose inhalers are not the most appropriate devices for nebulizing antibiotics and should be reserved for inhaled bronchodilators and corticosteroids.¹²

Jet nebulizers produce aerosol particles by superimposing an additional flow on the inspiratory flow that comes from the ventilator. The high-speed turbulent flow that generates the aerosol, however, also promotes particle impaction on ventilator circuits and proximal airways, thereby limiting deposition deep in the lung.

Ultrasonic nebulizers are appropriate for antibiotic nebulization for several reasons: They are generally equipped with a large reservoir; they generate particles whose mass median aerodynamic diameter is less than 5 μm through quartz vibrations; and the aerosol is entrained into the tracheobronchial tree by a low flow, independent of the inspiratory flow coming from the ventilator. The quartz vibration increases the temperature of the antibiotic solution and may alter the chemical structure of the antibiotic molecule.

Recently, vibrating plate nebulizers have been developed and seem appropriate for inhaled antibiotic therapy.¹⁷ The aerosol is generated by a ceramic vibrational element and a domed aperture plate, which has approximately 1000 tapered holes that are electroformed in a sheet. The antibiotic is placed in a reservoir above the domed aperture plate. Powered by an alternating current, the vibrational element expands and contracts causing the domed aperture plate to move upward and downward; this action causes a micropump effect that produces the aerosol.¹⁷ Particle size depends on the aperture diameter that can be changed by the manufacturer to optimize lung deposition. Vibrating plate nebulizers have several potential advantages over ultrasonic nebulizers: At the end of nebulization, the residual volume in the reservoir is negligible; the temperature of the antibiotic solution in the reservoir does not increase, thereby limiting the risk of altering the chemical structure of the antibiotic; the aerosol generation can be

synchronized with inspiration, minimizing aerosol waste during exhalation;¹⁸ and the aerosol can be delivered through an intratracheal catheter that can be inserted into a flexible bronchoscope.¹⁹ Both ultrasonic and vibrating plate nebulizers should be placed in parallel with the inspiratory limb before the Y-piece. The ventilator tubing between the nebulizer and the Y-piece serves as a reservoir containing the aerosol generated during the expiratory phase, which is entrained into the tracheobronchial tree during the next inspiration (bolus effect). Ultrasonic nebulizers should be placed 40 cm before the Y-piece to provide enough flexibility; vibrating plate nebulizers, which are less bulky, can be placed 10 cm before the Y-piece. When aerosol delivery is synchronized with inspiration, the vibrating plate nebulizer can be placed between the Y-piece and the proximal tip of the endotracheal tube, thus limiting extrapulmonary deposition.¹⁸ Limiting the nebulization period to a portion of the inspiratory phase, however, significantly increases the time of nebulization, which, in turn, reduces the maximal dose that can be nebulized.

If the nebulizer is placed distally to the endotracheal tube, instead of being directly connected to it, the inner diameter of the endotracheal tube does not markedly influence the efficiency of aerosol delivery.²⁰

Ventilator Modes and Settings

During mechanical ventilation, a laminar inspiratory flow provides better distal lung deposition of aerosol particles than a turbulent flow.²¹ Turbulence causes wall impaction of particles in the trachea and proximal bronchioles, thereby limiting antibiotic deposition in the lung parenchyma. As a consequence, specific ventilator settings aimed at limiting inspiratory flow turbulence should be implemented during the nebulization phase. In a patient fully adapted to the ventilator, a volume-controlled mode is preferred with the following settings: constant and low inspiratory flow, a minute ventilation limited to 6 L/min, a respiratory frequency of 12 breaths/min, an inspiratory-to-expiratory ratio of 50%, and an end-inspiratory pause that constitutes 20% of the duty cycle to facilitate settling of aerosol particles in the alveolar spaces.²² Decelerating flows should be avoided²³ as should patient triggering.²⁴ Inspiratory-to-expiratory ratios less than 50% should not be used because they do not provide enough time for settling of particles in the bronchoalveolar space.²⁵ These ventilator settings may not provide adequate CO₂ elimination in partially awake patients with acute bronchopneumonia. Consequently, dyssynchrony with the ventilator may result, generating turbulences and impaction of aerosolized particles in the ventilator circuits and the tracheobronchial tree. To provide adequate lung deposition, complete adaptation to the ventilator is required and can be achieved through a continuous infusion of propofol limited to the nebulization period.

Heat, Humidity, and Density of the Carrying Gas

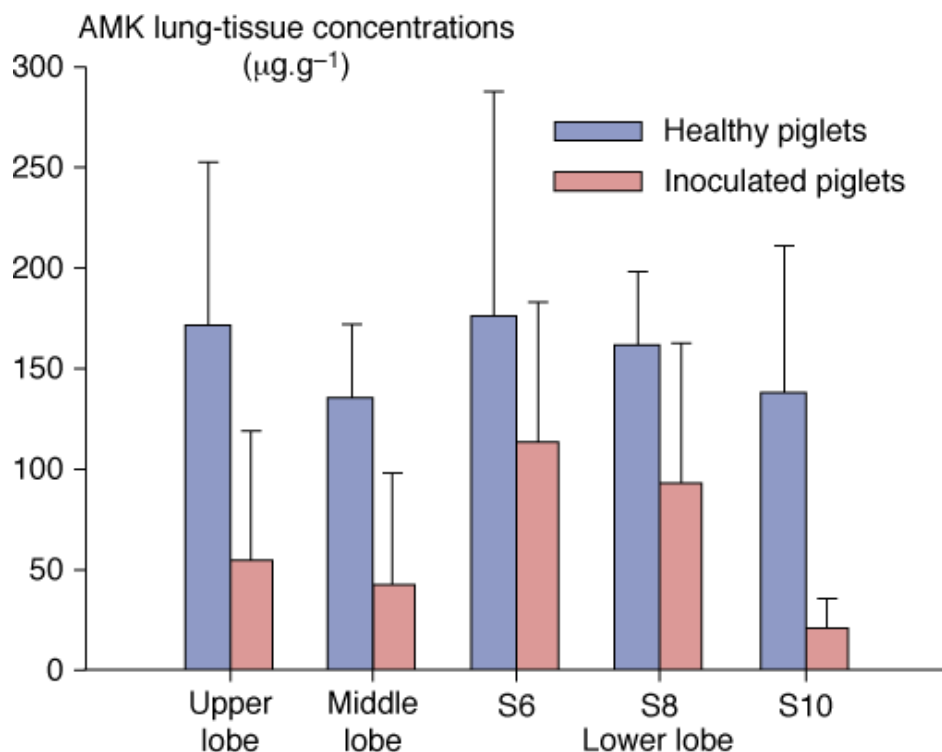
If the gas coming from the ventilator is heated and humidified, distal lung deposition of aerosol particles is markedly reduced.^{12,16} Two factors contribute to this undesirable effect: increase in the mass median aerodynamic diameter of aerosol particles²⁶ and increased deposition in ventilator circuits.²⁷ If the period of nebulization does not exceed 30 minutes, it is recommended to simply remove the heat and moisture exchanger; the aerosol will provide partial humidification of the inspired gas coming from the ventilator. If the period of nebulization exceeds 30 minutes, then a conventional humidifier should be inserted in the inspiratory limb to avoid damage to the tracheal and bronchial mucosa caused by prolonged administration of cold, dry inspiratory gas. As a result, the aerosol deposition might be reduced by 40%.¹²

Several in vitro studies have shown that reducing the density of the inspired gas, by replacing nitrogen with helium, increases lung deposition of aerosol particles.^{12,28,29} This beneficial effect likely results from helium-induced reduction in flow turbulence, limiting extrapulmonary deposition and tracheobronchial wall impaction of aerosol particles. On the other hand, operating the nebulizer with heliox (helium–oxygen gas mixture) reduces drug output and disposable mass.³⁰ Thus, it is recommended to operate the nebulizer with a nitrogen–oxygen mixture and to entrain aerosol particles with heliox to maximize lung deposition.¹² These in vitro studies have been confirmed in a recent experimental study performed in mechanically ventilated piglets with healthy lungs: Ceftazidime lung-tissue concentrations following aerosol administration were fivefold to 30-fold higher than after intravenous administration, and increased by 33% when heliox (65%/35%) was used as the operating gas of the ventilator.³¹ Unfortunately, this beneficial effect was not confirmed in animals with experimental *P. aeruginosa* bronchopneumonia; presumably, the loss of alveolar aeration resulting from bronchiolitis predominated over the helium-induced reduction in flow turbulence.³¹ At present, the routine use of heliox cannot be recommended for nebulizing antibiotics in ventilated patients with infected lungs.

Patient-Related Factors

Experimental studies performed in anesthetized piglets receiving prolonged mechanical ventilation have demonstrated that lung deposition of nebulized antibiotics is significantly greater in animals with healthy lungs than in animals with severe inoculation pneumonia.^{32–35} In addition, lung-tissue concentrations of antibiotics were homogeneously distributed in healthy animals, whereas they were heterogeneously distributed in infected animals. As shown in [Figure 64-2](#), dependent lung segments that were heavily infected by *Escherichia coli* had the lowest amikacin lung-tissue concentrations. The loss of lung aeration, the severity and extension of parenchymal infection, and the injury to the alveolar–capillary barrier are the factors that determine lung deposition of nebulized antibiotics.

Figure 64-2



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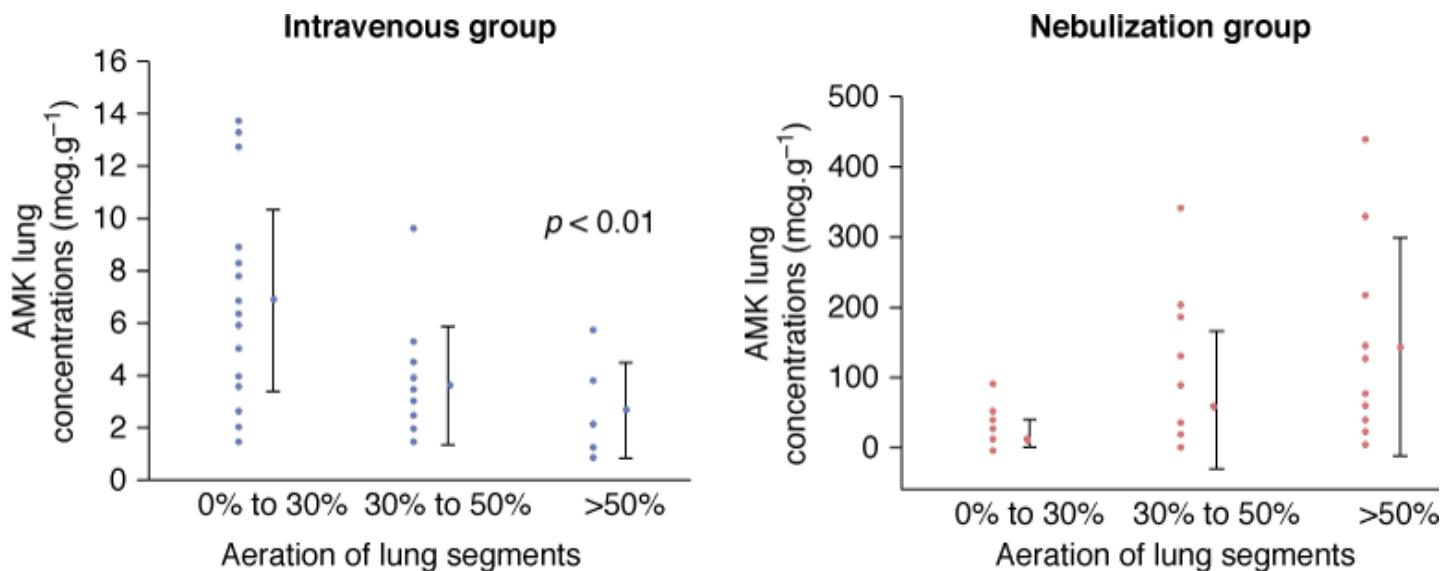
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Amikacin (AMK) lung-tissue concentrations (mean $\pm$ standard deviation) measured in different pulmonary lobes and segments 1 hour after a second daily nebulization of amikacin 45 mg/kg in mechanically ventilated piglets with healthy lungs (blue bars, $n = 5$) and in piglets with infected lungs (red bars, $n = 8$), inoculated 24 hours before the first administration with a solution of *Escherichia coli*. In the lower lobe, postmortem specimens were sampled from dependent (segments 6 and 10) and nondependent (segment 8) lung regions. (Used, with permission, from Goldstein et al.³³)

Lung Aeration

In patients with ventilator-associated bronchopneumonia, the obstruction of distal bronchioles by purulent plugs³⁶ may impair lung deposition of inhaled antibiotics, and result in lower pulmonary concentrations than those obtained in patients with healthy lungs. In anesthetized piglets receiving prolonged mechanical ventilation for a severe experimental *E. coli* bronchopneumonia, amikacin pulmonary tissue concentrations were three to thirty times higher after nebulization than after the intravenous administration of an equivalent dose.³³ As shown in Figure 64-3, the degree of lung aeration had the opposite effect in animals receiving intravenous amikacin and in animals receiving nebulized amikacin. The loss of lung aeration tended to increase amikacin tissue concentrations in the intravenous group, whereas the opposite effect was observed in the nebulization group. Very likely, the increased permeability of the alveolar-capillary barrier resulting from severe lung infection tends to promote intravenous amikacin penetration into the lung, whereas the multiple purulent plugs obstructing distal bronchioles tend to impair lung deposition of nebulized amikacin. It should be pointed out that despite the increased permeability of the alveolar-capillary barrier, amikacin lung-tissue concentrations remained hopelessly low after intravenous injection. From these experimental data, it is reasonable to hypothesize that ventilator settings aimed at recruiting nonaerated lung areas, such as positive end-expiratory pressure or recruitment maneuvers, may increase lung deposition of nebulized antibiotics.

Figure 64-3



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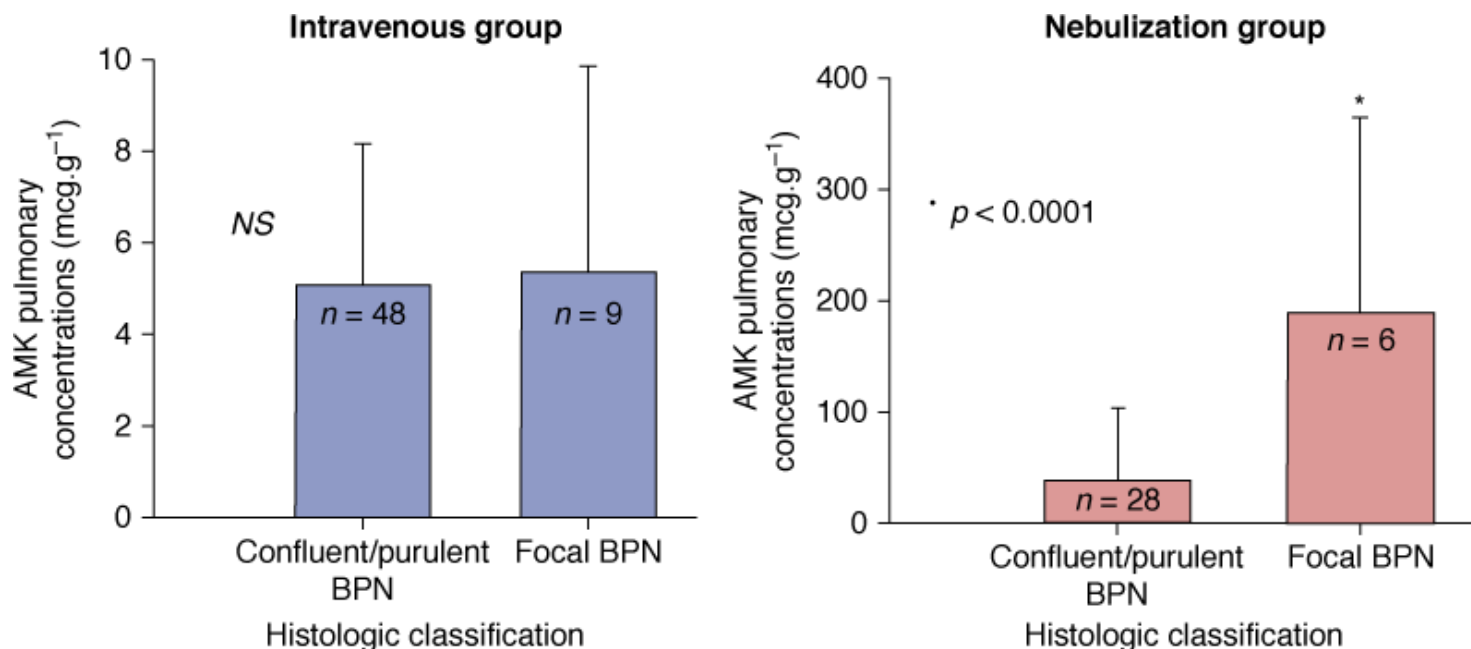
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Amikacin (AMK) pulmonary concentrations according to aeration of lung segments (each dot represents a single segment) measured in anesthetized piglets mechanically ventilated for a severe *Escherichia coli* bronchopneumonia. In the intravenous group, animals received an intravenous dose of amikacin (15 mg/kg). In the nebulization group, animals received an equivalent dose by the inhalation route (via ultrasonic nebulizer). Lung aeration was quantified on postmortem histologic samples using an image analyzer computerized system coupled to a high-resolution color camera and an optical microscope. (Used, with permission, from Elman et al.³⁸)

Severity and Extension of Bronchopneumonia

The extension and severity of human lung infection can be evaluated according to an histologic classification proposed in the early 1990s.^{36,37} Bronchiolitis is the immediate stage preceding lung infection. It is characterized by the proliferation of polymorphonuclear leukocytes within the bronchial lumen, forming purulent plugs, necrosis, and disruption of the bronchial mucosa. Neutrophilic infiltrates present in alveolar septa and distal bronchioles are characteristic of interstitial bronchopneumonia. The intense proliferation of polymorphonuclear leukocytes into surrounding alveoli characterizes foci of bronchopneumonia and decreases lung aeration. These foci of lung infection may extend to one or more pulmonary lobules resulting in confluent bronchopneumonia characterized by a severe aeration loss. Pulmonary abscess and purulent bronchopneumonia, the most severe forms of lung infection, are characterized by tissue necrosis, disruption of the normal lung architecture, vascular destruction, consolidation, and complete loss of aeration. In anesthetized piglets receiving prolonged mechanical ventilation for severe inoculation pneumonia, lung-tissue concentrations of nebulized amikacin, ceftazidime, or colistin were markedly higher in pulmonary segments with early stages of bronchopneumonia than in segments with confluent bronchopneumonia and lung abscess.^{34,35,38} As shown in Figure 64-4, such differences were not observed when the antibiotic was intravenously administered. These experimental data clearly support the administration of nebulized antibiotics in the early stages of ventilator-associated pneumonia.

Figure 64-4



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Amikacin (AMK) lung-tissue concentrations according to histologic stages of bronchopneumonia (BPN) characterizing lung segments (mean $\pm$ standard deviation), measured in anesthetized piglets mechanically ventilated for a severe *Escherichia coli* bronchopneumonia. In the intravenous group, animals received an intravenous dose of amikacin (15 mg/kg). In the nebulization group, animals received an equivalent dose by the inhalational route (via ultrasonic nebulizer). (Used, with permission, from Elman et al.³⁸)

Injury to the Alveolar–Capillary Barrier

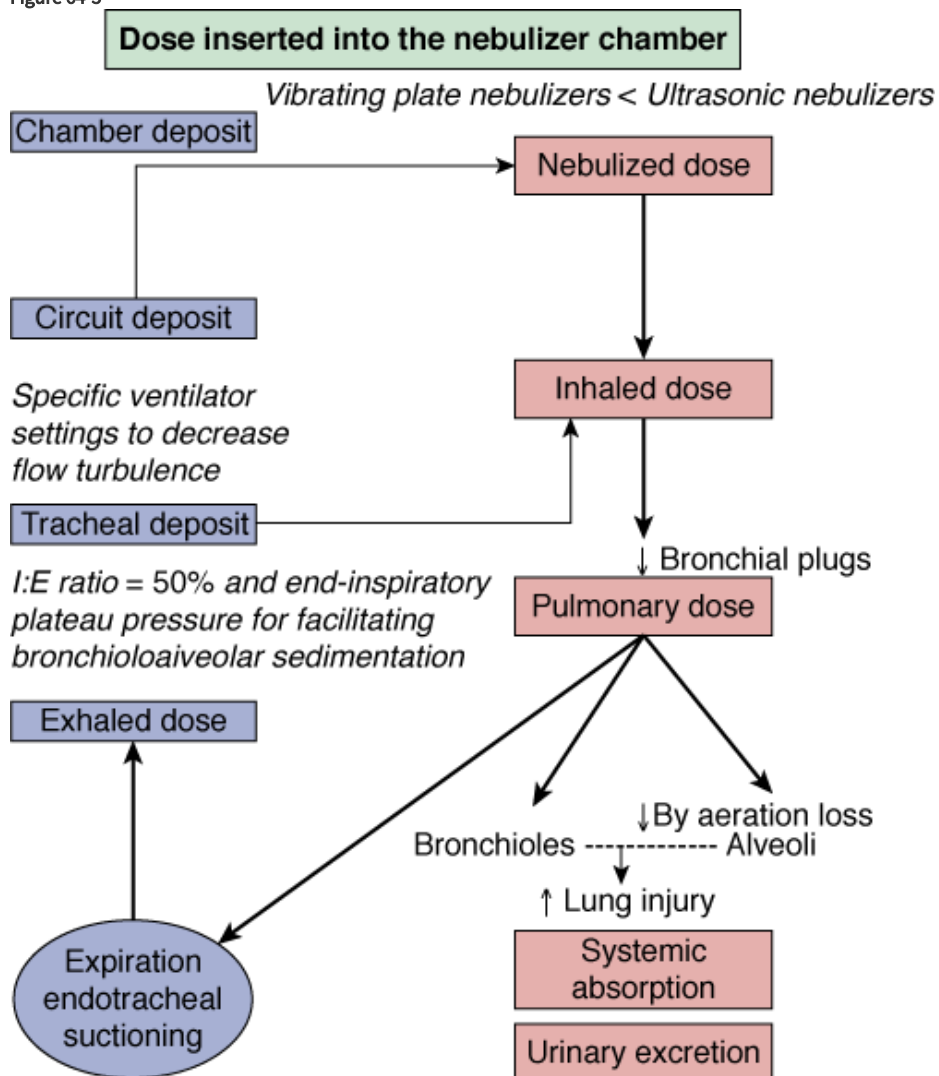
A critical factor influencing antibiotic lung deposition is the diffusion through physiologic barriers, such as the bronchial epithelium or vascular endothelium. The normal alveolar–capillary barrier offers high resistance to lung penetration of intravenous antibiotics and systemic diffusion of nebulized antibiotics.³² Lung infection, as with any type of lung injury, results in increased permeability of the alveolar–capillary barrier, which, in turn, facilitates the diffusion of nebulized antibiotics into the bloodstream.^{33,34} Experimentally, amikacin and ceftazidime plasma concentrations are no different after nebulization or intravenous administration in the presence of severe lung infection.^{33,34} In addition, systemic bioavailability is markedly increased. In other words, damage to the alveolar–capillary barrier, resulting from the infectious process, facilitates leakage of nebulized aminoglycosides and cephalosporins toward the systemic compartment, thereby decreasing the tissue concentration and increasing the risk of systemic toxicity. Interestingly, systemic diffusion of nebulized colistin was limited in piglets with inoculation *P. aeruginosa* pneumonia, suggesting a means of decreasing the systemic toxicity of polymyxins.³⁵

Efficiency of Aerosol Delivery

In the 1980s, lung deposition of aerosolized antibiotics was assessed using radiolabeled aerosols and gamma-scintigraphy.³⁹ In the early studies, mechanical ventilation was considered to limit the aerosol lung deposition to less than 5% of the dose placed in the nebulizer.⁴⁰ Over the following 15 years, improvements in aerosol technology, as well as an understanding of factors influencing lung deposition during mechanical ventilation, increased aerosol

delivery efficiency to 30% to 80%.^{15,16,20,32–35} Pneumatic jet nebulizers deliver less than 20% of the initial dose, because of high residual volume, massive deposition of aerosol particles in ventilator circuits and the endotracheal tube, and loss to atmosphere by the expiratory limb.²⁰ Ultrasonic nebulizers have increased the efficiency of aerosol delivery to 30% to 40%.^{32,33} The recent development of aerosol devices using vibrating plate technology, together with synchronizing aerosol delivery to the inspiratory phase, may even increase aerosol efficiency above 70%.^{20,34,35} Figure 64-5 illustrates the multiple factors influencing aerosol efficiency and lung deposition of nebulized antibiotics.

Figure 64-5



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The mechanisms by which the dose of antibiotic inserted into the nebulizer may differ markedly from the dose delivered to the distal lung. Patient-related factors and medical interventions can interfere with different steps in the nebulization process. (Thin arrows indicate limitation of lung deposition [arrow oriented down] or facilitation of systemic diffusion [arrow oriented up]).

Assessment of the Microbiologic Response

Methods for Assessing Microbiologic Response

In experimental animals, antibacterial efficiency of nebulized antibiotics can be directly assessed by measuring antibiotic lung-tissue concentrations and assessing quantitative bacteriology of postmortem lung-tissue samples.^{32–35} Concentrations measured from homogenized lung-tissue samples are representative of antibiotic present in interstitial and cell compartments. Most bacteria do not penetrate into cells and remain in the interstitial space, where antibiotics exert their bactericidal activity by binding to bacterial cell membrane. Therefore, antibiotic concentrations measured from homogenized lung biopsies underestimate “effective” interstitial concentrations, secondary to a dilution factor caused by intracellular components.⁴¹

In patients with ventilator-associated pneumonia, it can only be assessed indirectly as the regression of clinical, biologic, and radiologic signs of lung infection, and the disappearance of pathogens initially found in cultures of distal pulmonary samples. In ventilated critically ill patients, a number of confounding factors may hamper the clinical evaluation. The bronchial deposition of nebulized antibiotics may render cultures of distal samples falsely negative although the lung parenchyma is still positive. Critically ill patients frequently have several sites of infection, and persisting fever and biologic signs of infection may be related to a persisting extrapulmonary infection.

Antibiotic concentrations representative of the alveolar space can be assessed from a bronchoalveolar lavage. To obtain the antibiotic concentration representative of the epithelial lining fluid, the concentration measured in the aspirated fluid is corrected by a dilution factor, which is derived from the ratio between urea concentrations simultaneously measured in plasma and the bronchoalveolar lavage.⁴² Such measurements may fail to provide true representations of antibiotic concentrations at the site of lung infection because of confounding factors such as dilution errors or cell lysis.⁴² Of particular interest is a bronchoscopic microsampling method for measuring the antibiotic concentration in bronchial epithelial lining fluid.⁴³ It consists of introducing, through a bronchoscope positioned in the airways, an inner 1.9-mm polyester fiber rod probe, which immediately adsorbs fluid when placed on a bronchial wall for 10 seconds. Antibiotic concentrations are then measured from the probe without any dilution. To date, this attractive but expensive technique remains limited to research studies.

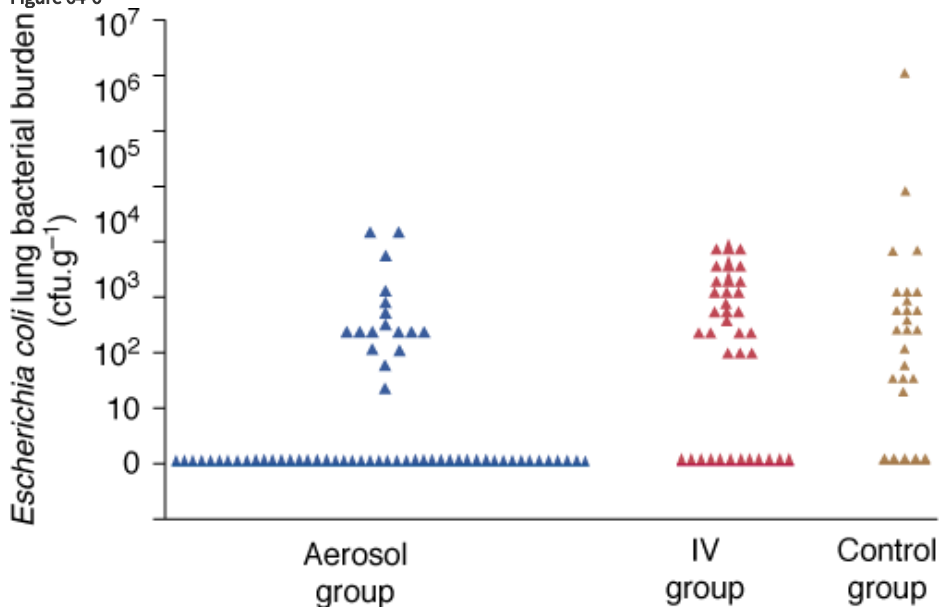
Microbiologic Response to Concentration-Dependent Antibiotics in Experimental Studies

Most experimental studies of nebulized antibiotics included aminoglycosides^{32,33,38,44–46} and polymyxins,³⁵ families of antibiotics exerting a concentration-dependent antibacterial effect.^{32,33,35,38,44–46} With concentration-dependent antibiotics, a single daily nebulization, ensuring high lung-tissue concentrations, is enough to achieve a bactericidal effect at the site of infection for a period of 12 to 24 hours. The postantibiotic effect prevents the regrowth of bacterial strains despite the antibiotic lung-tissue concentrations falling below the minimal inhibitory concentration. After several consecutive daily nebulizations of aminoglycosides, there is no time-dependent tissue accumulation.⁴⁷

In the late 1970s, experimental studies performed in nonventilated mice with *Klebsiella pneumoniae* bronchopneumonia had reported enhanced bacterial killing and higher survival rates when kanamycin was nebulized in comparison to intramuscular administration.^{44,45} In spontaneously breathing squirrel monkeys, these investigators demonstrated that nebulized kanamycin efficiently protected against *K. pneumoniae* bronchopneumonia induced by intratracheal bacterial instillation, whereas the intramuscular administration was not protective.⁴⁵ Antibiotic clearance curves indicated that nebulized kanamycin remained longer in the lungs and at

higher concentrations than intramuscular [kanamycin](#). In the early 1990s, a study performed in spontaneously breathing guinea pigs demonstrated that a combination of aerosolized and intramuscular [tobramycin](#) achieved slightly higher survival and total eradication of *P. aeruginosa* compared to nebulized or intramuscular [tobramycin](#) alone.⁴⁶ Ten years later, an experimental study looked at the antibacterial efficiency of nebulized amikacin in anesthetized piglets ventilated for a severe *E. coli* bronchopneumonia.³³ Twenty-four hours after a massive bronchial inoculation, ventilated animals received equivalent doses of amikacin, either by ultrasonic nebulization or intravenously. Because of a 60% extrapulmonary deposition, 45 mg/kg were nebulized in a single dose and 15 mg/kg administered intravenously. The animals received a second dose after 24 additional hours of mechanical ventilation and were killed 1 hour later, and five subpleural specimens were excised from the upper, middle, and lower lobes. Amikacin lung-tissue peak concentrations were threefold to 30-fold higher after nebulization than after intravenous administration. As shown in [Figure 64-6](#), after two nebulizations and 25 hours of treatment, 71% of lung segments were sterile, whereas cultures of lung segments were comparable in nontreated and intravenously treated animals. In 2010, the antibacterial efficiency of nebulized colistin was assessed in anesthetized piglets ventilated for severe *P. aeruginosa* bronchopneumonia.³⁵ Twenty-four hours after a massive bronchial inoculation, ventilated animals received equivalent doses of colistin, either by nebulization or intravenously. Because of a 60% extrapulmonary deposition, 16 mg/kg were nebulized in two daily doses and 10 mg/kg administered intravenously in three doses. The animals were killed 1 hour after the third aerosol and the fourth intravenous administration, and five subpleural specimens were excised from the upper, middle, and lower lobes. Colistin lung-tissue concentrations were not detectable following intravenous administration and onefold to 12-fold higher than minimal inhibitory concentrations after nebulization. After three nebulizations and 25 hours of treatment, 67% of lung segments were sterile, whereas 88% of lung segments were positive in nontreated animals and 72% in intravenously treated animals.

Figure 64-6



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Lung bacterial burden of *Escherichia coli* (expressed in colony-forming units per gram [cfu/g] of lung tissue) in lung segments of anesthetized piglets mechanically ventilated for severe *E. coli* bronchopneumonia. Postmortem tissue samples were collected 1 hour after the second aerosol or intravenous (IV) dose of amikacin, or 48 hours after the bacterial inoculation in the untreated control group. Each *triangle* refers to a single lung segment. The lung bacterial

burden is significantly lower in the aerosol group as compared with the intravenous or control groups. (Used, with permission, from Goldstein et al.³³)

Microbiologic Response to Time-Dependent Antibiotics in Experimental Studies

With time-dependent antibiotics, such as cephalosporins, tissue concentrations have to be maintained permanently above minimal inhibitory concentrations to provide a bactericidal effect. Therefore, aerosols have to be repeated several times a day.

An experimental study looked at the lung-tissue concentrations of ceftazidime administered to anesthetized piglets ventilated for a severe *P. aeruginosa* bronchopneumonia.³¹ Twenty-four hours after a massive bacterial inoculation, animals received equivalent doses of ceftazidime either by ultrasonic nebulization or intravenously. Because of a 30% extrapulmonary deposition, 50 mg/kg were nebulized in a single dose and 33 mg/kg administered intravenously. The animals were sacrificed 1 hour later, and five subpleural specimens were excised from the upper, middle, and lower lobes. Ceftazidime lung-tissue concentrations following nebulization were fivefold to 30-fold higher than after intravenous administration.³¹ In a second study performed in the same experimental model, animals received either nebulized or intravenous ceftazidime for 24 hours.³⁴ Twenty-four hours after a massive *P. aeruginosa* bronchial inoculation, ventilated animals received equivalent doses of ceftazidime, either by nebulization or intravenously. Because of a 60% extrapulmonary deposition, 200 mg/kg were nebulized in eight daily aerosols of 25 mg/kg each, and 90 mg/kg was administered by continuous intravenous infusion. The animals were killed 3 hours after the eighth aerosol and 24 hours after initiation of intravenous administration, and five subpleural specimens were excised from the upper, middle, and lower lobes. Ceftazidime trough lung-tissue concentrations were fourfold higher after nebulization compared to continuous intravenous administration. After nine nebulizations and 25 hours of treatment, 80% of lung segments were sterile, whereas 90% of lung segments were positive in nontreated animals and 70% in intravenously treated animals.

Clinical Response in Human Studies

The experimental studies reported above clearly suggest that aminoglycosides, polymyxins, and cephalosporins have a higher bactericidal efficiency when administered by nebulization as compared to the parenteral route. Beneficial effects of nebulization and endotracheal administration of antibiotics have been repeatedly reported in spontaneously breathing patients with cystic fibrosis.⁴⁸ Several human studies have also demonstrated that the endotracheal administration or nebulization of aminoglycosides and polymyxins may prevent bronchial infection and ventilator-associated pneumonia in ventilated critically ill patients.^{49–52} The risk of encouraging resistive strains, however, has limited this prophylactic approach.⁵³ Two studies performed in ventilated patients treated by intravenous antibiotics demonstrated that the addition of endotracheal **tobramycin** is useful for eradicating the pathogens that cause gram-negative pneumonia.^{54,55} In 1992, a comparative pharmacokinetic study, performed in ventilated patients with nosocomial pneumonia, reported high antibiotic bronchial concentrations following the nebulization or the endotracheal administration of 1 g of ceftazidime.⁵⁶ In addition, the minimal inhibitory concentrations for 90% of the most important pathogens responsible for nosocomial infections were exceeded by concentrations in bronchial secretions for up to 12 hours after intravenous infusion and for up to 24 hours after endotracheal and aerosol administration.⁵⁶

More than 40 years ago, aerosols of colistin were successfully administered to spontaneously breathing patients with pulmonary suppuration.⁵⁷ In the early 1970s, two spontaneously breathing patients with *P. aeruginosa* pneumonia were treated with inhaled polymyxin B, but aerosols had to be stopped because of airway obstruction.⁵⁸ Over the last decade, several investigator groups have reported microbiologic response following nebulization of antibiotics to critically ill patients with ventilator-associated pneumonia.^{59–70} Most of these studies were retrospective and concerned patients treated by nebulized colistin for ventilator-associated pneumonia caused by multidrug-resistant pathogens.^{59,60,62,63,66–69} Three studies were prospective^{64,65,70} and two were randomized.^{65,70} Most investigators reported “beneficial” effects, either resulting from a combination of nebulized and intravenous antibiotics^{59,60,65,65} or from nebulized antibiotics alone.^{61–63,67–70} Except for a prospective randomized study published in 2011,⁷⁰ ventilator settings were not optimized in these studies and the rationale for choosing the aerosol dose was not given. Clinical response was considered beneficial when aerosol antibiotics were efficient in treating patients whose lungs were infected with multidrug-resistant microorganisms.^{60,62,64,65,67,69,70}

In a matched case-control study, aerosolized plus intravenous colistin was not more efficient than intravenous colistin alone for treating ventilator-associated pneumonia caused by multidrug-resistant pathogens.⁶⁸ In a double-blind, randomized, placebo-controlled study performed in forty-five critically ill patients with ventilator-associated tracheobronchitis, aerosolized [gentamicin](#) or vancomycin administered over the course of 14 days and combined with systemic antibiotics were associated with several benefits⁶⁵: better resolution of ventilator-associated pneumonia, less bacterial resistance, reduced used of systemic antibiotics, and facilitation of weaning from mechanical ventilation. Several limitations, however, preclude drawing firm conclusions from the study: The small number of patients included in the aerosol group ($n = 19$), the small doses used in the aerosol group ([gentamicin](#) = 180 mg/day and vancomycin = 180 mg/day), the lack of optimization of ventilator settings during nebulization, and the heterogeneity of clinical and bacterial criteria for determining efficiency of antimicrobial therapy. In a randomized, controlled, phase II trial performed in forty critically ill patients with ventilator-associated pneumonia caused by *P. aeruginosa*, aerosolized ceftazidime and amikacin were compared to intravenous ceftazidime and amikacin.⁷⁰ In both arms, antimicrobial therapy was administered over the course of 7 days. Resolution of ventilator-associated pneumonia was assessed by objective criteria based on eradication of *P. aeruginosa* from bronchoalveolar lavage, resolution of clinical signs of sepsis, and lung reaeration on computed tomography. During aerosol administration, ventilator settings were optimized to minimize extrapulmonary deposition and intravenous and nebulized doses were determined so as to deliver comparable amounts of ceftazidime and amikacin in the trachea and pulmonary artery. Aerosol and intravenous antibiotics had a similar efficiency for treating ventilator-associated pneumonia caused by *P. aeruginosa* sensitive to ceftazidime and amikacin. Unlike systemic antibiotics, aerosolized antibiotics were efficient against intermediate strains and they reduced the emergence of resistive *P. aeruginosa*. Because of the small number of patients included, no conclusion could be drawn about the potential benefit of aerosol antibiotics on emergence of resistive strains, duration of mechanical ventilation, length of stay in the intensive care unit, and mortality. Multicenter randomized trials are required to determine the clinical impact and the indications of nebulized antibiotics, and more specifically their efficiency in ventilator-associated pneumonia caused by multidrug-resistant microorganisms. Apart from potential benefit in patients with ventilator-associated pneumonia, nebulized antibiotics may also prevent microbial biofilm formation on the endotracheal tube internal surface,⁷⁴ thereby suppressing a reservoir of infecting microorganisms for the deep lung.⁷

Important Unknowns and Issues to Be Resolved

Two major issues must be taken into consideration before nebulized antibiotics can be recommended as an alternative to intravenous anti-infectious therapy in critically ill patients.

Side Effects of Nebulized Antibiotics

Very few complications have been described in patients receiving inhaled antibiotics. Bronchoconstriction, chest tightness, and apnea are the main adverse effects of colistin aerosols.^{72,74,75} Hypoxemia may result from aerosol nebulization of any drug, particularly in patients with acute lung injury or acute respiratory distress syndrome. With aminoglycosides, the most common adverse effects are tinnitus, hoarseness, voice alteration, wheezing, cough, dyspnea, and bronchospasm.⁷³ Nebulized cephalosporins or other antibiotics are usually well tolerated. Cough, nasal congestion, wheezing, taste disturbances, and chest tightness have been reported.⁷³ Obstruction of the expiratory filter caused by repetitive aerosols of cephalosporin have been reported in ventilated patients with a risk of increased airway pressure and cardiac arrest.⁷⁰

Another serious potential risk of nebulized antibiotics is the emergence of multiresistant pathogens. Some old studies reported the emergence of resistant strains complicating the intratracheal administration of gentamycin, sisomicin, or colistin for preventing and treating ventilator-associated pneumonia.^{49,52,53,76} Other experimental⁵¹ and clinical⁷⁷ studies did not find an increase in the incidence of resistant pathogens when polymyxin B or colistin were endotracheally administered to prevent ventilator-associated pneumonia. Interestingly, nebulized antibiotics for treating ventilator-associated pneumonia seem to reduce the emergence of multidrug-resistant strains.^{65,70}

Determination of the Aerosol Dose

Different rationales can be adopted for optimizing the aerosol dose. To deliver comparable amounts of antibiotics in the trachea and pulmonary artery, aerosol dose can be determined as intravenous dose plus extrapulmonary deposition. Extrapulmonary deposition can be assessed by rinsing nebulization chamber and the inspiratory and expiratory circuits with saline and measuring antibiotic concentration in the recovered fluid.^{31–35,38,70,78} Another rationale is to define the “best” aerosol dose as the dose providing systemic concentrations in the range of those obtained after intravenous administration.⁷⁹ Such a rationale requires repetitive systemic dosages and appears difficult to implement in clinical practice. Studies are required to assess the impact of these different strategies on clinical efficiency of inhaled antibiotics.

Guidelines for Inhaled Antibiotic Therapy

It is important to recognize that inhaled antibiotics for treating ventilator-associated pneumonia differ in several ways from inhaled bronchodilator therapy for treating bronchospasm in ventilated patients. Nebulized antibiotics must penetrate into the distal lung, whereas bronchodilators only need to reach the bronchial tree. Because antibiotic tissue concentrations in the infected lung parenchyma should largely exceed minimal inhibitory concentrations of causative pathogens, optimizing dosage, an unresolved issue, is more critical for inhaled antibiotic therapy than for inhaled bronchodilator therapy. Great attention must be paid to the period of nebulization: Specific ventilator settings should be used to limit flow turbulence, and a short-acting sedative may be required to avoid flow triggering and

dyssynchrony with the ventilator. Figures 64-7 and 64-8 summarize guidelines for delivering nebulized antibiotics during mechanical ventilation.

Figure 64-7

Use ultrasonic or vibrating plate nebulizers, producing aerosols whose particles have a mass median aerodynamic diameter $< 5 \mu\text{m}$.

Remove heat and moisture exchanger and conventional humidifier and stop humidification during the period of nebulization.

Place the nebulizer on the inspiratory limb, 20 cm from the Y piece.

Determine in vitro the extrapulmonary deposition in the ventilator circuits using ventilator settings applied during the nebulization period:

The amount of antibiotic deposited into inspiratory and expiratory circuits should be measured after lavage of each part of the circuit with a known volume of water.

Determine the daily dose to be placed in the nebulizer chamber:

If the aminoglycoside is administered exclusively by nebulization, the dose should be calculated as the intravenous dose + extrapulmonary deposition. If the aminoglycoside is concomitantly intravenously administered, then the determination of the appropriate dosage is difficult. Trough plasma concentrations should be monitored daily in order to avoid systemic accumulation.

If colistin is administered exclusively by nebulization, the dose should range between 6 and 15 million international units/day. If it is also intravenously administered, then the determination of the appropriate dosage is difficult. Trough plasma concentrations should be monitored daily in order to avoid systemic accumulation.

Determine the interval between each nebulization:

*For aminoglycosides, a single daily nebulization.
For colistin, 3 daily nebulizations (every 8 hours).*

Use specific ventilatory settings during the nebulization period, which should not exceed 30 minutes (for details see Fig. 64-8).

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Guidelines for inhaled antibiotic therapy in ventilated patients.

Figure 64-8

Use a controlled mode of mechanical ventilation

with the following ventilator settings:

- *Constant inspiratory flow*
- *Tidal volume of 6 – 8 mL/kg*
- *Respiratory frequency 12 bpm*
- *Inspiratory to expiratory ratio 1:1*
- *Inspiratory plateau pressure 20%*
- *Remove any humidification system*
- *Optimize alveolar recruitment*

Avoid assisted modes of mechanical ventilation:

Where the patient triggers flow during spontaneous inspiratory efforts

**Avoid discoordination of the patient
with the ventilator:**

If necessary, provide sedation with a continuous infusion of propofol during the nebulization period

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Specific ventilator settings for inhaled antibiotic therapy in ventilated patients.

Conclusions and the Future

Although inhaled antibiotic therapy has been episodically used in ventilated patients for more than 40 years, convincing clinical data are still unavailable to support its routine use. The recent understanding of physical and physiologic factors influencing lung deposition of aerosolized antibiotics together with an impressive growth of new technologies fuel a renewed interest for treating ventilator-associated pneumonia with nebulized antibiotics. In 2012, administration of inhaled antibiotics to ventilated patients should be judged a potentially efficient therapy for treating ventilator-associated pneumonia caused by resistant microorganisms. In critically ill patients with ventilator-associated pneumonia caused by sensitive bacteria, well-conducted phase II trials suggest that inhaled antibiotic therapy is at least equivalent to intravenous antibiotics and may decrease the emergence of resistant strains.^{65,70} Multicenter randomized studies should be set up to assess the impact of inhaled antibiotic therapy on mortality, morbidity, antibiotic consumption, and costs in critically ill patients with ventilator-associated pneumonia.

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